

**Installation Standard
For
POLYETHYLENE (PE) COLD WATER BUILDING SUPPLY AND YARD PIPING**

IAPMO IS 7-90^{e1}

This standard shall govern the installation of polyethylene (PE) cold water building supply and yard piping. (See Section 604.1 of the Uniform Plumbing Code and Section 609.2 of this standard for allowable location and pressure). Installation, material and inspection shall comply with the current edition of the Uniform Plumbing Code published by the International Association of Plumbing and Mechanical Officials, and shall also comply with this standard.

Note: *The following sections of the Uniform Plumbing Code apply to polyethylene piping.*

202.0	Definition of PE
Table 14-1	Fittings (and fasteners) for joining PE pipe IAPMO PS25-84 Plastic insert fittings for Polyethylene (PE) Plastic Pipe ASTM D2609-96a PE Pipe ASTM D2239-96a
310.0	Workmanship
313.0	Protection of Piping, Materials and Structures
314.0	Supporting in the Ground
315.0	Backfilling
316.2.3	Connection to other materials
Chapter 6	Water Distribution
604.1	Location
609.1	Depth of piping
609.3	
103.5.4.2 & 103.5.3.3	Testing

Chapter 2 DEFINITIONS

Chapter 14	
ASTM	American Society for Testing and Materials
IAPMO	International Association of Plumbing and Mechanical Officials
NSF	National Sanitation Foundation
PS	Material and Property Standard published by IAPMO
UPC	Uniform Plumbing Code published by IAPMO

301.1 Minimum Standards

301.1.1 Material. Material shall conform to the appropriate standard in Table 14-1. See note ahead of Chapter 2 of this standard.

301.1.1.1 Pipe. PE pipe is plastic and black.

301.1.1.2 Fittings. Fittings are copper alloy or nylon barbed insert fittings.

301.1.2 Markings

301.1.2.1 Piping. Pipe shall be marked at intervals of not more than five (5) feet (1524 mm) with at least the following:

- (a) Manufacturer's name or trademark;
- (b) ASTM D2239;
- (c) Pipe size;
- (d) PE 3406 or PE 3408;
- (e) Pressure rating of SDR 7 - PE 3406 (160 psi/1.10 mPa) and PE 3408 (200 psi/1.38 mPa);
- (f) NSF seal or marking of other acceptable agency; and
- (g) Pipe listed by IAPMO that is covered by this standard shall be labeled with the UPC logo to show compliance with this standard.

301.1.2.2 Fittings. Fittings shall be marked with at least the following:

- (a) Manufacturer's name or trademark;
- (b) Size; and
- (c) Fittings listed by IAPMO that are covered by this standard shall be labeled with the UPC logo to show compliance with this standard.

301.1.2.3 Bands. Bands shall be marked with at least the following:

- (a) Manufacturer's name or trademark;
- (b) Model;
- (c) Stainless steel, Series 300; and
- (d) Bands listed by IAPMO that are covered by this standard shall be labeled with the UPC logo to show compliance with this standard.

301.1.3 Position of Markings. The identifying markings on pipe, and fittings shall be visible for inspection without moving materials.

313.0 Protection of Piping

313.1 Storage. Unprotected pipe should not be stored in direct sunlight. The pipe shall be stored in a way to protect it from mechanical damage (slitting, puncturing, etc.).

313.2 Thermal Expansion. The pipe shall be snaked in the trench bottom with enough slack to provide for thermal expansion and contraction. The normal slack created by

	residual coiling is generally sufficient for this purpose. If, however, the pipe has been allowed to straighten before it is placed in the trench, six (6) inches (152 mm) per one hundred (100) feet (30480 mm) of length shall be allowed for this purpose.	It shall not be installed within or under any building or structure or mobile home or commercial coach or parts thereof. The term "building or structure or parts thereof" shall include structures such as porches and steps, whether covered or uncovered, roofed porte-cocheres, roofed patios, carports, covered walks, covered driveways and similar structures or appurtenances.
313.3	Exposed Piping. Vertical piping may extend a maximum of twenty-four (24) inches (610 mm) above grade when located on the exterior of the building or structure and protected from mechanical damage to the satisfaction of the Administrative Authority. Where exposed to sunlight the pipe shall be wrapped with at least 0.040 in. (1.02 mm) of tape.	604.2 Harmful Materials. Polyethylene that has been in contact with gasoline, lubricating oil or aromatic compounds, shall not be installed.
315.0	Trenching and Cover	609.0 Installation
315.1	Trench bottoms shall be uniformly graded and shall be of either undisturbed soil or shall consist of a layer or layers of compacted backfill so that minimum settlement will take place.	609.1 Pipe. Kinked pipe shall not be used. PE pipe shall not be flared.
		609.2 Fittings. Compression type couplings and fittings shall be used only when installing one and one-half (1-1/2) inch (38 mm) and larger pipe. Stiffeners that extend beyond the clamp or nut shall not be used.
316.1	Joints	609.3 Bends. Changes in direction may be made by bends. The installed radius of pipe curvature shall be not less than thirty (30) pipe diameters, or the coil radius when bending with the coil. Coiled pipe shall not be bent beyond straight. Bends shall not be permitted closer than ten (10) pipe diameters of any fitting or valve.
316.1.1	Procedure. Polyethylene pipe joints shall be made as follows (see Section 301.1.1): Step 1. Pipe shall be cut square, using a cutter designed for plastic pipe, and chamfer ends to remove sharp edges. Step 2. Place two strap-type stainless steel bands over the pipe. Step 3. Check that fittings are properly sized for pipe, as tubing fittings are not of proper size. Step 4. Force the end of the pipe over the barbed insert fittings, making contact with the fitting shoulder (the end of the pipe may be softened by placing in hot water). Step 5. Position the clamps 180° apart and tighten evenly, so as to make a leak-proof joint.	609.4 Maximum Working Pressure. Working pressure shall not exceed 160 psi (1.10 mPa).
		609.5 Identification. A label shall be fastened to the main electric meter panel stating "This structure has a non-metallic water service".
316.1.2	Other Joints. Polyethylene pipe shall not be threaded. Joints made with adhesives or "solvent cementing" techniques are prohibited.	610.0 Sizing
604.0	Materials	610.1 Piping shall be sized in accordance with UPC Section 610.0. When UPC Appendix A is applicable, use UPC Chart A-4 (Copper Tubing Type L). Flow velocity shall not exceed 8 fps (2.4 m/s).
604.1	Location. Polyethylene piping shall be installed only outside the foundation of any building or structure or parts thereof. It shall be buried in the ground for its entire length except vertical piping may be extended above grade per Section 313.3.	

ADOPTED: 1968

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